

Vectors

I. Vector Basics

Defn: A **vector**, $\mathbf{v}$, is a mathematical object that has both _____ and _____

Notation:

- Vectors will be written either in bold or with an arrow: $\mathbf{v}, \vec{v}$.
- A vector, $\vec{v}$, in $\mathbf{R}^2$ ($\vec{v} \in \mathbf{R}^2$) can be represented by _____ coordinates: $\vec{v} =$ _____
- A vector, $\vec{v}$, in $\mathbf{R}^3$ ($\vec{v} \in \mathbf{R}^3$) can be represented by _____ coordinates: $\vec{v} =$ _____
- The special unit vectors $\mathbf{i} = [1,0,0]$, $\mathbf{j} = [0,1,0]$, and $\mathbf{k} = [0,0,1]$ form a basis for $\mathbf{R}^3$, so we can write $\vec{v} = [v_1, v_2, v_3]$ as _____

Ex:

- $[10, 14], [0, 1], [-1, 2]$ _____ and $[10, 4] =$ _____
- $[1, 1, -1], [0, 0, 0], [-10, 12, -8]$ _____ and $[-10, 12, -8] =$ _____

Thm: Let $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$ be vectors. Then

$$\mathbf{a} = \mathbf{b} \text{ iff } \underline{\hspace{1cm}}$$

Ex: $[2, 0, 2d + c] = [c, 0, 1] \implies \underline{\hspace{1cm}}$

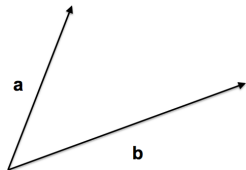
II. Vector Addition

Vector Operation 1: Addition

Let $\mathbf{a} = [a_1, a_2, a_3]$ and $\mathbf{b} = [b_1, b_2, b_3]$

Then

$$\mathbf{a} + \mathbf{b} = [a_1, a_2, a_3] + [b_1, b_2, b_3] =$$



Note:

- $\mathbf{a}, \mathbf{b} \in \mathbf{R}^3 \implies$
- $\mathbf{a} + \mathbf{b}$ is known as the
the

of $\mathbf{a}$ and $\mathbf{b}$. (e.g. If $\mathbf{a}$ and $\mathbf{b}$ were forces then $\mathbf{a} + \mathbf{b}$ would be

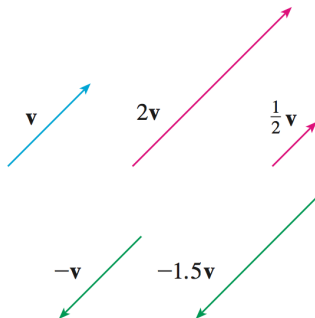
III. Scalar Multiplication

Property: Scalar Multiplication of Vectors

Let $\mathbf{v} = [v_1, v_2, v_3] \in \mathbf{R}^3$ and $c \in \mathbf{R}$ (scalar)

Then

$$c\mathbf{v} = c[v_1, v_2, v_3] =$$



Note:

- Scalar multiplication can $\quad$, $\quad$, or $\quad$ a vector.

IV. Length(Norm) of a Vector

Defn: Let $\mathbf{v} = [v_1, v_2]$ and $\mathbf{w} = [w_1, w_2, w_3]$. Then the length of $\mathbf{v}$ and $\mathbf{w}$ (denoted $|\mathbf{v}|$ and $|\mathbf{w}|$) is given by

$$|\mathbf{v}| = \quad \text{and} \quad |\mathbf{w}| =$$

Ex: Find the length of $\mathbf{w} = [1, -2, 4]$

$$|\mathbf{w}| =$$

Note:

- $\mathbf{w} \in \mathbf{R}^3$ (vector), but $|\mathbf{w}| \in$

Defn: $\mathbf{v}$ is a unit vector if

Ex: Find a unit vector, $\mathbf{u}$, in the same direction as $\mathbf{w}$ above.

$$\mathbf{u} =$$